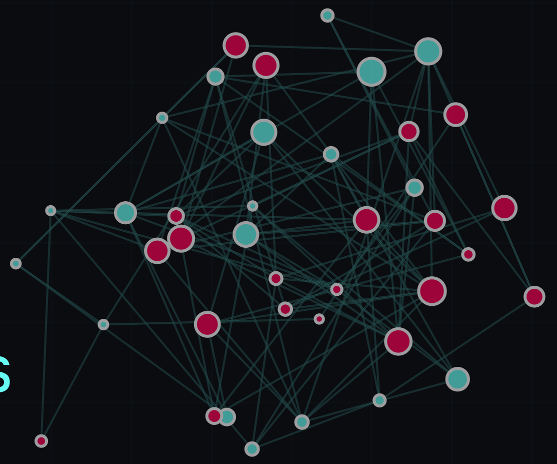
CHAPTER 07

Classification: Sorting the World



From Spam Filters to Medical Diagnosis

▸ CAI1001C | Artificial Intelligence Thinking

▸ Professor Carlos Marquez

▸ Miami Dade College

Mission Briefing

LEARNING OBJECTIVES

- Differentiate Classification from Regression
- Train and interpret a k-NN classifier
- Explain how the value of K impacts the model
- Train and visualize a Decision Tree
- Compare the performance of two different classifiers

Two Types of Prediction

SUPERVISED LEARNING TASKS

Regression

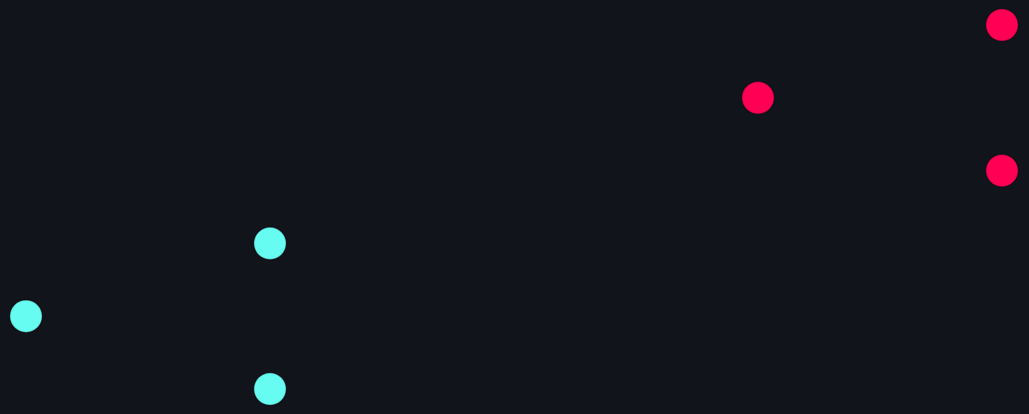
Asks: "How much?"

Output: A Number

(e.g., $25,000)

Classification

Asks: "Which one?"



Output: A Category

(e.g., Spam / Not Spam)

k-Nearest Neighbors (k-NN)

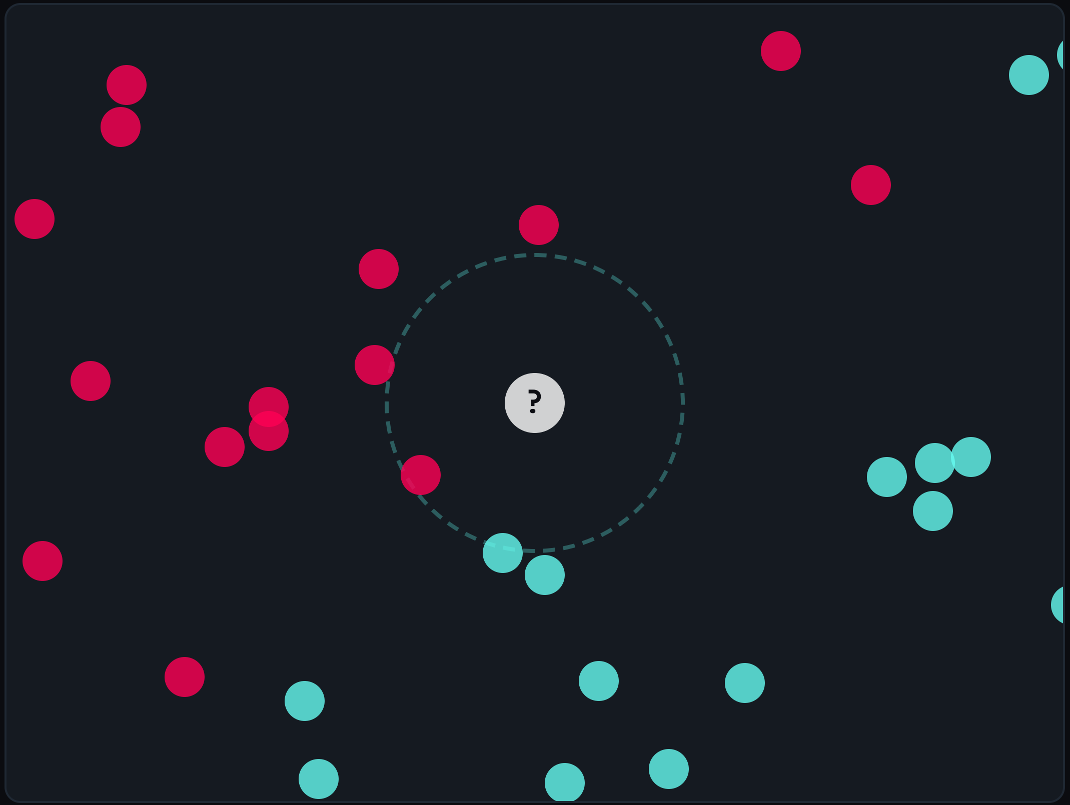
THE INTUITION

How do we classify an unknown data point?

Simple: We look at its closest neighbors.

The Core Principle:
Things that are close to each other in data space likely belong to the same category.

“ If it walks like a duck and quacks like a duck...



The k-NN Workflow

HOW THE ALGORITHM THINKS

1 Choose K

Select the number of neighbors to consult (e.g., K=3 or K=5).

2 Measure Distance

Calculate the distance between the new point and every training example.

3 Majority Vote

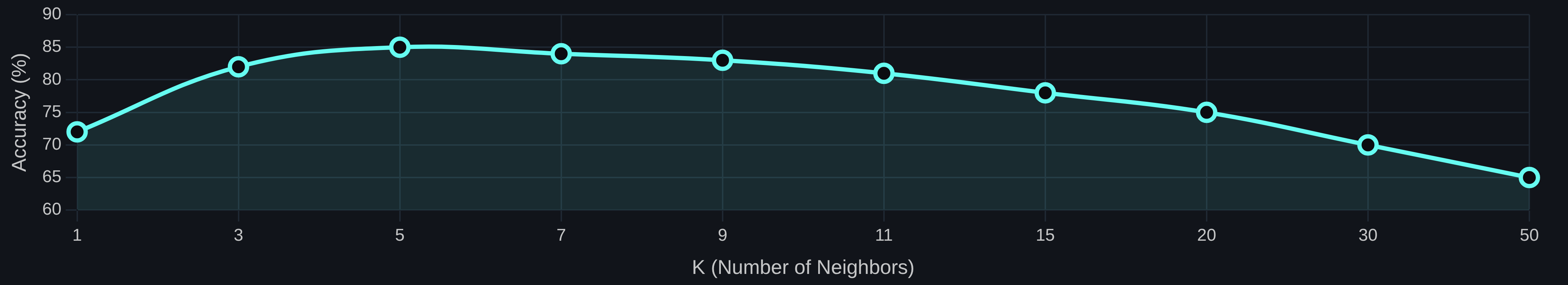
Count the labels of the K nearest neighbors. The winner takes all.

CRITICAL REQUIREMENT: FEATURE SCALING

k-NN uses distance, so large numbers (e.g., Salary) overpower small ones (e.g., Age). **Always normalize your data first!**

Finding the Right-Sized Neighborhood

THE GOLDILOCKS PROBLEM



Too Small (K=1)

Sensitive to noise. One outlier can flip the prediction.
Risk: Overfitting

Just Right (K=5)

Balances sensitivity and stability. Captures the pattern.
Goal: Generalization

Too Large (K=100)

Averages out everything. Misses local details.
Risk: Underfitting

The Flowchart Approach

DECISION TREES

The Strategy

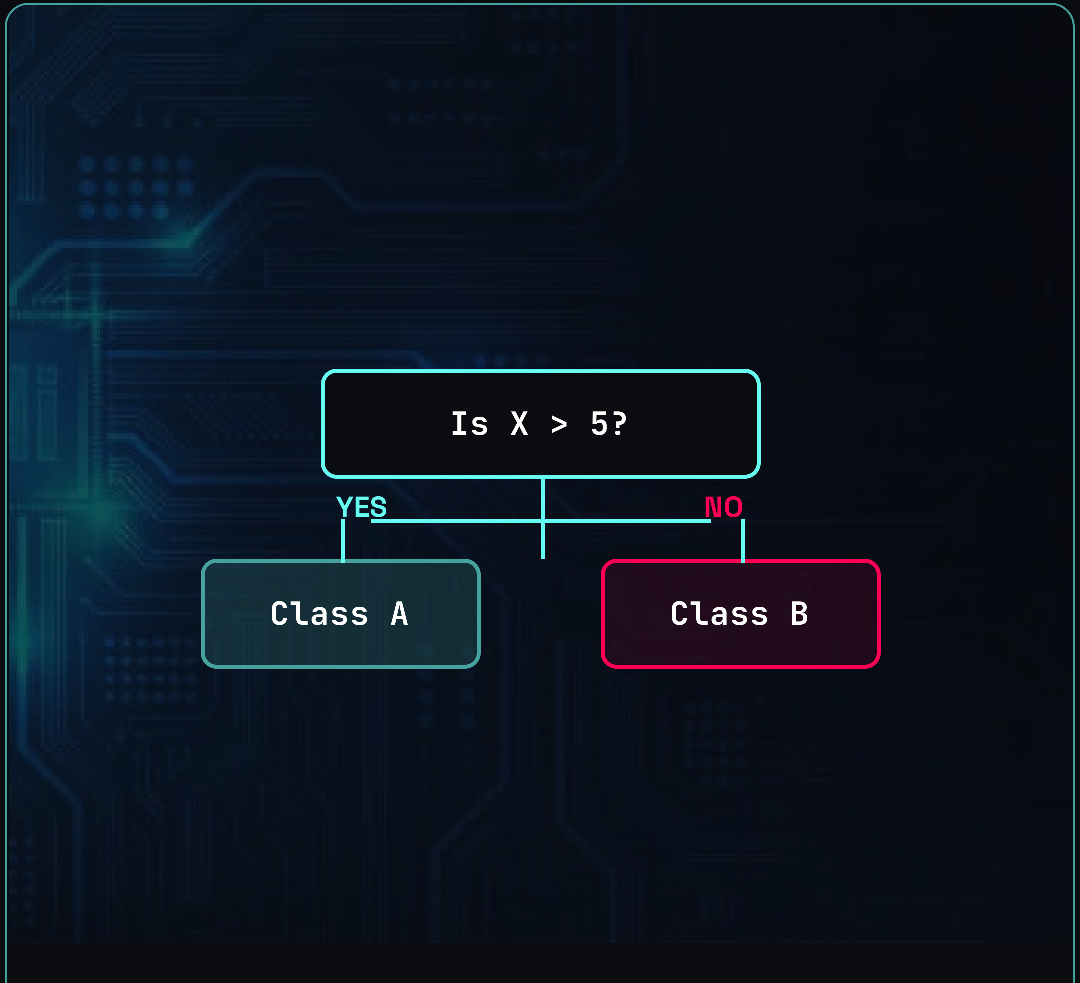
Instead of measuring distance, the model asks a sequence of **Yes/No questions** about the features.

The Goal

Split the data into smaller, purer groups until a decision can be made.

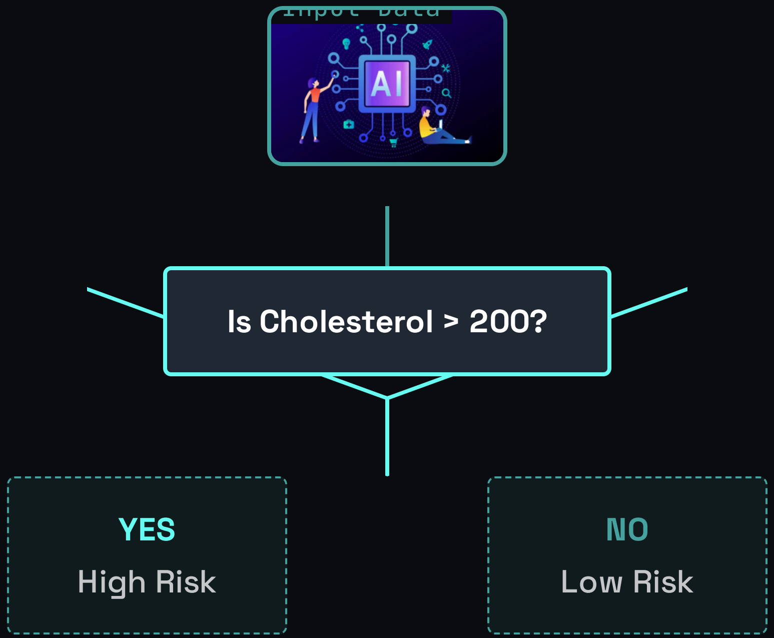
THINK ABOUT IT:

It's like a game of **20 Questions.** You don't guess random objects; you ask "Is it alive?" then "Is it an animal?" to narrow it down.



The Logic of Splits

ANATOMY OF A DECISION TREE



Node (The Question)

A decision point that tests one feature.
"Is age > 50?"

Branch (The Path)

The Yes/No path the data follows based on the answer.

Leaf (The Answer)

The final endpoint where a prediction is made.

KEY CONCEPT: INFORMATION GAIN

The algorithm calculates which question best separates the groups. The "purest" split wins.

Algorithm Tradeoffs

HEAD-TO-HEAD COMPARISON

FEATURE	K-NEAREST NEIGHBORS	DECISION TREE
Strategy	Distance & Voting (Geometric)	Splitting Rules (Logical)
Clarity	Low (Black Box)	High (White Box)
Speed	Slow Prediction (Lazy Learner)	Fast Prediction (Eager Learner)
Key Risk	Sensitive to Scale & Outliers	Prone to Overfitting (Memorization)

The Python Implementation

SCIKIT-LEARN PIPELINE

k-Nearest Neighbors

SCALING REQUIRED

```python
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler

# 1. Scale the features (CRITICAL!)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 2. Initialize Model (K=5)
knn = KNeighborsClassifier(n_neighbors=5)

# 3. Train
knn.fit(X_scaled, y)
```

Decision Tree

NO SCALING NEEDED

```python
from sklearn.tree import DecisionTreeClassifier


# 1. No scaling required
# Trees handle raw data well


# 2. Initialize Model
dt = DecisionTreeClassifier(max_depth=3)

# 3. Train
dt.fit(X, y)
```

Performance on a Real Dataset

CASE STUDY: HEART DISEASE PREDICTION

Model Accuracy

Winner: k-NN

Top Predictors (Decision Tree)

The Hidden Bias Problem

WHEN "ACCURATE" ISN'T FAIR

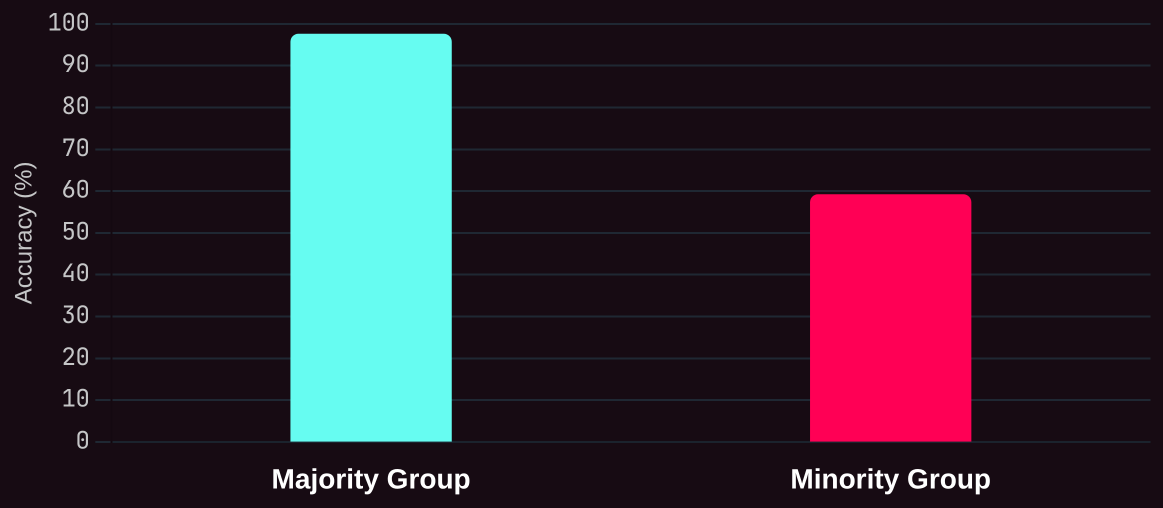
95%

Overall Accuracy

"The model looks great! It gets an A grade on the test set."

But averages hide the truth.

THE REALITY: BREAKDOWN BY GROUP



The Danger: If the training data is 90% Group A, the model learns to ignore Group B to maximize its score.

Mission Debrief

KEY TAKEAWAYS

Classification vs. Regression

We predict **categories** (labels), not continuous numbers.

k-Nearest Neighbors (k-NN)

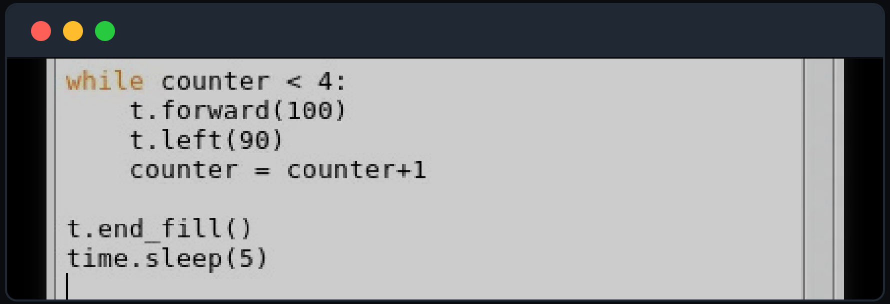
Classifies by majority vote. Simple but sensitive to the value of K and requires **feature scaling.**

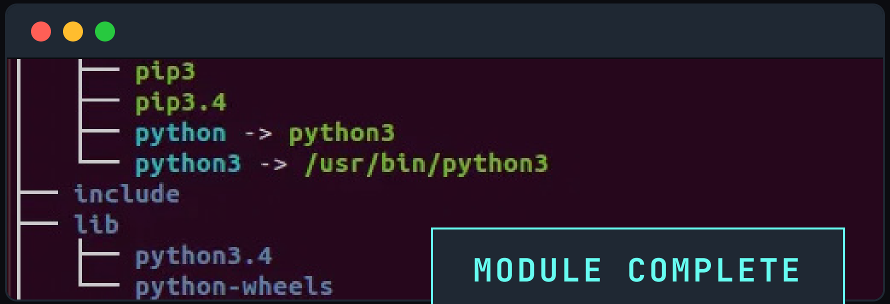
Decision Trees

Uses a flowchart of Yes/No questions. Highly **interpretable** but prone to overfitting if too deep.

Ethics Check

Accuracy isn't enough. We must check for bias across different groups.





MODULE COMPLETE

What's Next?

CHAPTER 8: ADVANCED CLASSIFICATION

Linear Classifiers

Logistic Regression and the power of drawing straight lines through data.

SVMs

Support Vector Machines: Finding the widest possible margin between groups.

>_ Prepare for higher dimensions...